ZERO DIMENSIONAL LOCALLY ALGEBRAIC SCHEMES ARE DISJOINT UNIONS OF SPECTRA OF LOCAL ARTINIAN ALGEBRAS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and ends with the finiteness of X over Spec(k); no step is referred to the reader.

Editorial note. The counted spans carry 4 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-algebraic-scheme-dim-0 is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Noether normalization for a zero dimensional algebra over a field. Algebra, Lemma algebra-lemma-Noether-normalization, invoked at proof lines 3748-3750. Supplies the finite injection k to A, hence finite k-dimension of A.
- A finite dimensional algebra over a field is Artinian and a product of Artinian local rings. Algebra, Lemmas algebra-lemma-finite-dimensional-algebra and algebra-lemma-artinian-finite-length, invoked at proof lines 3751-3755. Supplies the product decomposition of A into Artinian local rings.
- A finite disjoint union of affine schemes is affine. Schemes, Lemma schemes-lemma-disjoint-union-affines, invoked at proof lines 3772-3773. Supplies affineness of a finite disjoint union of spectra in the algebraic case.

Source reference. Original lemma 00OY: sources/stacks-chow/tags.txt, source lines 912–912. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00JB: sources/stacks-chow/tags.txt, source lines 709–709. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00J6: sources/stacks-chow/tags.txt, source lines 704–704. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 01I5: sources/stacks-chow/tags.txt, source lines 1966–1966. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be a locally algebraic k-scheme of dimension 0. Then X is a disjoint union of spectra of local Artinian k-algebras A with $\dim_k(A) < \infty$. If X is an algebraic k-scheme of dimension 0, then in addition X is affine and the morphism $X \to \mathrm{Spec}(k)$ is finite.*

Proof. Let X be a locally algebraic k-scheme of dimension 0. Let $U = \mathrm{Spec}(A) \subset X$ be an affine open subscheme. Since $\dim(X) = 0$ we see that $\dim(A) = 0$. By Noether normalization, see Algebra, Lemma 00OY we see that there exists a finite injection $k \to A$, i.e., $\dim_k(A) < \infty$. Hence A is Artinian, see Algebra, Lemma 00J6. This implies that $A = A_1 \times \ldots \times A_r$ is a product of finitely many Artinian local rings, see Algebra, Lemma 00JB. Of course $\dim_k(A_i) < \infty$ for each i as the sum of these dimensions equals $\dim_k(A)$.

The arguments above show that X has an open covering whose members are finite discrete topological spaces. Hence X is a discrete topological space. It follows that X is isomorphic to the disjoint union of its connected components each of which is a singleton. Since a singleton scheme is affine we conclude (by the results of the paragraph above) that each of these singletons is the spectrum of a local Artinian k-algebra A with $\dim_k(A) < \infty$.

Finally, if X is an algebraic k-scheme of dimension 0, then X is quasi-compact hence is a finite disjoint union $X = \mathrm{Spec}(A_1) \amalg \ldots \amalg \mathrm{Spec}(A_r)$ hence affine (see Schemes, Lemma 01I5) and we have seen the finiteness of $X \to \mathrm{Spec}(k)$ in the first paragraph of the proof. □